Module 19: Implement Site-to-Site IPsec VPNs with CLI

Networking Security v1.0
(NETSEC)

Module Objectives

Module Title: Implement Site-to-Site IPsec VPNs with CLI

Module Objective: Configure a site-to-site IPsec VPN, with pre-shared key authentication, using CLI.

Topic Title	Topic Objective
Configure a Site-to-Site IPsec VPN	Describe IPsec negotiation and the five steps of IPsec configuration.
ISAKMP Policy	Use the correct commands to configure an ISAKMP policy.
IPsec Policy	Use the correct commands to configure the IPsec policy.
Crypto Map	Use the correct command to configure and apply a Crypto map.
IPsec VPN	Configure the IPsec VPN.

19.1 Configure a Site-to-Site IPsec VPN

IPsec Negotiation

IPsec negotiation to establish a VPN involves five steps, which include IKE Phase 1 and Phase 2:

1. An ISAKMP tunnel is initiated when host A sends “interesting” traffic to host B. Traffic is considered interesting when it travels between the peers and meets the criteria that are defined in an ACL.
2. IKE Phase 1 begins. The peers negotiate the ISAKMP SA policy. When the peers agree on the policy and are authenticated, a secure tunnel is created.
3. IKE Phase 2 begins. The IPsec peers use the authenticated secure tunnel to negotiate the IPsec SA policy. The negotiation of the shared policy determines how the IPsec tunnel is established.
4. The IPsec tunnel is created, and data is transferred between the IPsec peers based on the IPsec SAs.
5. The IPsec tunnel terminates when the IPsec SAs are manually deleted, or when their lifetime expires.

Site-to-Site IPsec VPN Topology

Implementing a site-to-site VPN requires configuring settings for both IKE Phase 1 and Phase 2. In the phase 1 configuration, the two sites are configured with the necessary ISAKMP security associations to ensure that an ISAKMP tunnel can be created. In the phase 2 configuration, the two sites are configured with the IPsec security associations to ensure that an IPsec tunnel is created within the ISAKMP tunnel. Both tunnels will be created only when interesting traffic is detected.

The topology in the figure for XYZCORP will be used in this section to demonstrate a site-to-site IPsec VPN implementation.

Site-to-Site IPsec VPN Topology (Cont.)

Both routers are configured with IP addressing and static routing. An extended ping on R1 verifies that routing between the LANs is operational.

```
R1# show run
<output omitted>
!
interface GigabitEthernet0/0
 ip address 10.0.1.1 255.255.255.0
!
interface Serial0/0/0
 ip address 172.30.2.1 255.255.255.0
!
ip route 192.168.1.0 255.255.255.0 Serial0/0/0

!=========================================

R2# show run

<output omitted>
!
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
!
interface Serial0/0/0
 ip address 172.30.2.2 255.255.255.0
!
ip route 10.0.1.0 255.255.255.0 Serial0/0/0
!
```

```
R1# ping 192.168.1.1 source 10.0.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
Packet sent with a source address of 10.0.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R1#
```

IPsec VPN Configuration Tasks

Security Policy Requirements

All XYZCORP VPNs should be implemented using the following security policy:

- Encrypt traffic with AES 256 and SHA.
- Authenticate with PSK.
- Exchange keys with DH group 14.
- ISAKMP tunnel lifetime is 1 hour.
- IPsec tunnel uses ESP with a 15-minute lifetime.

Configuration Tasks:

The configuration tasks required to meet this policy are:

Task 1: Configure the ISAKMP Policy for IKE Phase 1
Task 2: Configure the IPsec Policy for IPsec Phase 2
Task 3: Configure a Crypto Map for the IPsec Policy
Task 4: Apply the IPsec Policy
Task 5: Verify that the IPsec Tunnel is Operational

Existing ACL Configurations

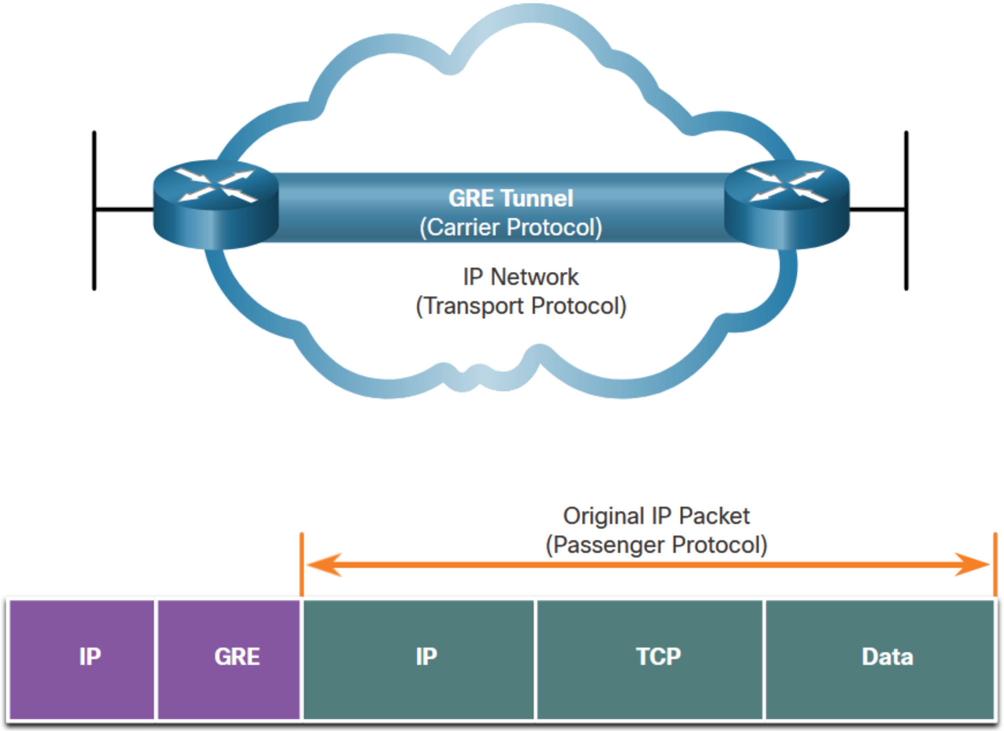
Prior to implementing a site-to-site IPsec VPN, ensure that the existing ACLs do not block traffic necessary for IPsec negotiations. The ACL command syntax to permit ISAKMP, ESP, and AH traffic is shown here.

```
Router(config)# ip access-list extended name
Router(config-ext-nacl)# permit udp source wildcard destination wildcard eq isakmp
Router(config-ext-nacl)# permit esp source wildcard destination wildcard
Router(config-ext-nacl)# permit ahp source wildcard destination wildcard
```

```
R1(config)# ip access-list extended INBOUND
R1(config-ext-nacl)# permit ip 192.168.1.0 0.0.0.255  10.0.1.0 0.0.0.255
R1(config-ext-nacl)# permit icmp host 172.30.2.2 host 172.30.2.1
R1(config-ext-nacl)# permit udp host 172.30.2.2 host 172.30.2.1 eq isakmp
R1(config-ext-nacl)# permit esp host 172.30.2.2 host 172.30.2.1
R1(config-ext-nacl)# permit ahp host 172.30.2.2 host 172.30.2.1
R1(config-ext-nacl)# deny ip any any
R1(config-ext-nacl)# exit
R1(config)# interface serial0/0/0
R1(config-if)# ip access-group INBOUND in
```

Handling Broadcast and Multicast Traffic

The XYZCORP topology uses static routing, so there is no multicast or broadcast traffic that needs to be routed through the tunnel. But what if XYZCORP decided to implement EIGRP or OSPF? To enable routing protocol traffic, the peers in a site-to-site IPsec VPN implementation would need to be configured with a Generic Routing Encapsulation (GRE) tunnel for the multicast traffic.



19.2 ISAKMP Policy

The Default ISAKMP Policies

The Cisco IOS comes with default ISAKMP policies already in place. To view the default policies, enter the **show crypto isakmp default policy** command.

R1 has seven default ISAKMP policies ranging from the most secure (policy 65507) to the least secure (policy 65514). If no other policy has been defined by the administrator, R1 will attempt to use the most secure default policy. If R2 has a matching policy, then R1 and R2 can successfully negotiate the IKE Phase 1 ISAKMP tunnel without any configuration by the administrator.

The Default ISAKMP Policies (Cont.)

In this example, none of the default policies match the security policy for XYZCORP. A new ISAKMP policy will have to be configured.

```
R1# show crypto isakmp default policy

Default IKE policy
Default protection suite of priority 65507
          encryption algorithm:   AES - Advanced Encryption Standard (128 bit keys).
          hash algorithm:         Secure Hash Standard
          authentication method:  Rivest-Shamir-Adleman Signature
          Diffie-Hellman group:   #5 (1536 bit)
          lifetime:               86400 seconds, no volume limit
Default protection suite of priority 65508
          encryption algorithm:   AES - Advanced Encryption Standard (128 bit keys).
          hash algorithm:         Secure Hash Standard
          authentication method:  Pre-Shared Key
          Diffie-Hellman group:   #5 (1536 bit)
          lifetime:               86400 seconds, no volume limit
Default protection suite of priority 65509
          encryption algorithm:   AES - Advanced Encryption Standard (128 bit keys).
          hash algorithm:         Message Digest 5
          authentication method:  Rivest-Shamir-Adleman Signature
          Diffie-Hellman group:   #5 (1536 bit)
          lifetime:               86400 seconds, no volume limit
Default protection suite of priority 65510
          encryption algorithm:   AES - Advanced Encryption Standard (128 bit keys).
          hash algorithm:         Message Digest 5
          authentication method:  Pre-Shared Key
          Diffie-Hellman group:   #5 (1536 bit)
          lifetime:               86400 seconds, no volume limit
Default protection suite of priority 65511
          encryption algorithm:   Three key triple DES
          hash algorithm:         Secure Hash Standard
          authentication method:  Rivest-Shamir-Adleman Signature
          Diffie-Hellman group:   #2 (1024 bit)
          lifetime:               86400 seconds, no volume limit
```

```
Default protection suite of priority 65512
          encryption algorithm:   Three key triple DES
          hash algorithm:         Secure Hash Standard
          authentication method:  Pre-Shared Key
          Diffie-Hellman group:   #2 (1024 bit)
          lifetime:               86400 seconds, no volume limit
Default protection suite of priority 65513
          encryption algorithm:   Three key triple DES
          hash algorithm:         Message Digest 5
          authentication method:  Rivest-Shamir-Adleman Signature
          Diffie-Hellman group:   #2 (1024 bit)
          lifetime:               86400 seconds, no volume limit
Default protection suite of priority 65514
          encryption algorithm:   Three key triple DES
          hash algorithm:         Message Digest 5
          authentication method:  Pre-Shared Key
          Diffie-Hellman group:   #2 (1024 bit)
          lifetime:               86400 seconds, no volume limit
```


Syntax to Configure a New ISAKMP Policy

To configure a new ISAKMP policy, use the **crypto isakmp policy** command. The only argument for the command is to set a priority for the policy (from 1 to 10000). Peers will attempt to negotiate using the policy with the lowest number (highest priority).

When in ISAKMP policy configuration mode, the SAs for the IKE Phase 1 tunnel can be configured. Use the mnemonic **HAGLE** to remember the five SAs to configure:

- **H**ash
- **A**uthentication
- **G**roup
- **L**ifetime
- **E**ncryption

```
R1(config)# crypto isakmp policy ?
  <1-1000> Priority of protection suite

R1(config)# crypto isakmp policy 1
R1(config-isakmp)# ?

ISAKMP commands:
  authentication   Set authentication method for protection suite
  default
Set a command to its defaults
  encryption   Set encryption algorithm for protection suite
  exit   Exit from ISAKMP protection suite configuration mode
  group   Set the Diffie-Hellman group
  hash   Set hash algorithm for protection suite
  lifetime   Set lifetime for ISAKMP security association
  no   Negate a command or set its defaults]]>
```

ISAKMP Policy Configuration

To meet the security policy requirements for XYZCORP, configure the ISAKMP policy with the following SAs:

- Hash is SHA
- Authentication is pre-shared key
- Group is 14
- Lifetime is 3600 seconds
- Encryption is AES

```
R1(config)# crypto isakmp policy 1
R1(config-isakmp)# hash sha
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# group 24
R1(config-isakmp)# lifetime 3600
R1(config-isakmp)# encryption aes 256
R1(config-isakmp)# end
R1# show crypto isakmp policy
Global IKE policy
Protection suite of priority 1
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys).
        hash algorithm: Secure Hash Standard
        authentication method: Pre-Shared Key
        Diffie-Hellman group: #24 (2048 bit, 256 bit subgroup)
        lifetime: 3600 seconds, no volume limit
R1#
```

Configuring a Pre-Shared Key

The pre-shared key command syntax is as follows:

```
Router(config)# crypto isakmp key keystring address peer-address
Router(config)# crypto isakmp key keystring hostname peer-hostname
```

XYZCORP uses the key phrase **cisco12345** and the IP address of the peer:

```
R1# conf t
R1(config)# crypto isakmp key cisco12345 address 172.30.2.2
R1(config)#
```

```
R2# conf t
R2(config)# crypto isakmp key cisco12345 address 172.30.2.1
R2(config)#
```

19.3 IPsec Policy

Define Interesting Traffic

Although the ISAKMP policy for the IKE Phase 1 tunnel is configured, the tunnel does not yet exist. This is verified with the **show crypto isakmp sa** command.

```
R1# show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status

IPv6 Crypto ISAKMP SA

R1#
```

To define interesting traffic, configure each router with an ACL to permit traffic from the local LAN to the remote LAN. The ACL will be used in the crypto map configuration to specify what traffic will trigger the start of IKE Phase 1.

```
R1# conf t
R1(config)# access-list 101 permit ip 10.0.1.0 0.0.0.255 192.168.1.0 0.0.0.255
R1(config)#
```

```
R2# conf t
R2(config)# access-list 102 permit ip 192.168.1.0 0.0.0.255 10.0.1.0 0.0.0.255
R2(config)#
```


Configure IPsec Transform Set

The next step is to configure the transform set, a set of encryption and hashing algorithms that will be used to transform the data sent through the IPsec tunnel.

Configure a transform set using the **crypto ipsec transform-set** command. First, specify a name for the transform set (R1-R2, in the example). The encryption and hashing algorithm can then be configured in either order.

```
R1(config)# crypto ipsec transform-set ?
  WORD  Transform set tag

R1(config)# crypto ipsec transform-set R1-R2 ?
  ah-md5-hmac      AH-HMAC-MD5 transform
  ah-sha-hmac      AH-HMAC-SHA transform
  ah-sha256-hmac   AH-HMAC-SHA256 transform
  ah-sha384-hmac   AH-HMAC-SHA384 transform
  ah-sha512-hmac   AH-HMAC-SHA512 transform
  comp-lzs         IP Compression using the LZS compression algorithm
  esp-3des         ESP transform using 3DES(EDE) cipher (168 bits)
  esp-aes          ESP transform using AES cipher
  esp-des          ESP transform using DES cipher (56 bits)
  esp-gcm          ESP transform using GCM cipher
  esp-gmac         ESP transform using GMAC cipher
  esp-md5-hmac     ESP transform using HMAC-MD5 auth
  esp-null         ESP transform w/o cipher
  esp-seal         ESP transform using SEAL cipher (160 bits)
  esp-sha-hmac     ESP transform using HMAC-SHA auth
  esp-sha256-hmac  ESP transform using HMAC-SHA256 auth
  esp-sha384-hmac  ESP transform using HMAC-SHA384 auth
  esp-sha512-hmac  ESP transform using HMAC-SHA512 auth
```

```
R1(config)# crypto ipsec transform-set R1-R2 esp-aes esp-sha-hmac
R1(config)#
```

```
R2(config)# crypto ipsec transform-set R1-R2 esp-aes esp-sha-hmac
R2(config)#
```

19.4 Crypto Map

Syntax to Configure a Crypto Map

Now that the interesting traffic is defined, and an IPsec transform set is configured, it is time to bind those configurations with the rest of the IPsec policy in a crypto map. The available configurations for a crypto map entry when you are in crypto map configuration mode are shown below. Although the **ipsec-manual** option is shown, its use is beyond the scope of this course.

```
Router(config)# crypto map map-name seq-num { ipsec-isakmp | ipsec-manual }
```

Parameter	Description
map-name	Identifies the crypto map set.
seq-num	Sequence number you assign to the crypto map entry. Use the **crypto map** *map-name seq-num* command without any keyword to modify the existing crypto map entry or profile.
ipsec-isakmp	Indicates that IKE will be used to establish the IPsec for protecting the traffic specified by this crypto map entry.
ipsec-manual	Indicates that IKE will not be used to establish the IPsec SAs for protecting the traffic specified by this crypto map entry.

Syntax to Configure a Crypto Map (Cont.)

The available configurations for a crypto map entry when you are in crypto map configuration mode are shown below. The map name is **R1-R2_MAP**, and the sequence number is **10**.

```
R1(config)# crypto map R1-R2_MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)# ?
Crypto Map configuration commands:
  default          Set a command to its defaults
  description      Description of the crypto map statement policy
  dialer           Dialer related commands
  disable          Disable this crypto-map-statement.
  exit             Exit from crypto map configuration mode
  match            Match values.
  no               Negate a command or set its defaults
  qos              Quality of Service related commands
  reverse-route    Reverse Route Injection.
  set              Set values for encryption/decryption
```

Crypto Map Configuration

To finish the configuration to meet the IPsec security policy for XYZCORP, complete the following:

Step 1. Bind the ACL and the transform set to the map.
Step 2. Specify the peer's IP address.
Step 3. Configure the DH group.
Step 4. Configure the IPsec tunnel lifetime.

```
R1(config)# crypto map R1-R2_MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)# match address 101
R1(config-crypto-map)# set transform-set R1-R2
R1(config-crypto-map)# set peer 172.30.2.2
R1(config-crypto-map)# set pfs group24
R1(config-crypto-map)# set security-association lifetime seconds 900
R1(config-crypto-map)# exit
R1(config)#
```

```
R2(config)# crypto map R1-R2_MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R2(config-crypto-map)# match address 102
R2(config-crypto-map)# set transform-set R1-R2
R2(config-crypto-map)# set peer 172.30.2.1
R2(config-crypto-map)# set pfs group24
R2(config-crypto-map)# set security-association lifetime seconds 900
R2(config-crypto-map)# exit
R2(config)#
```

Crypto Map Configuration (Cont.)

Use the **show crypto map** command to verify the crypto map configuration, as shown in here. All the required SAs should be in place.

```
R1# show crypto map
Crypto Map IPv4 "R1-R2_MAP" 10 ipsec-isakmp
    Peer = 172.30.2.2
    Extended IP access list 101
        access-list 101 permit ip 10.0.1.0 0.0.0.255 192.168.1.0 0.0.0.255
    Security association lifetime: 4608000 kilobytes/900 seconds
    Responder-Only (Y/N): N
    PFS (Y/N): Y
    DH group:  group24
    Mixed-mode : Disabled
    Transform sets={
            R1-R2:  { esp-aes esp-sha-hmac  } ,
    }
    Interfaces using crypto map R1-R2_MAP:

R1#
```


Apply and Verify the Crypto Map

To apply the crypto map, Use the **crypto map** *map-name* interface configuration command to apply the crypto map.

Use the **show crypto map** to verify the crypto map is not applied to the interface.

```
R1(config)# interface serial0/0/0
R1(config-if)# crypto map R1-R2_MAP
R1(config-if)#
*Mar 19 19:36:36.273: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)# end
R1# show crypto map
Crypto Map IPv4 "R1-R2_MAP" 10 ipsec-isakmp
          Peer = 172.30.2.2
          Extended IP access list 101
              access-list 101 permit ip 10.0.1.0 0.0.0.255 192.168.1.0 0.0.0.255
          Security association lifetime: 4608000 kilobytes/900 seconds
          Responder-Only (Y/N): N
          PFS (Y/N): Y
          DH group:  group24
          Mixed-mode : Disabled
          Transform sets={
                  R1-R2:  { esp-aes esp-sha-hmac  } ,
          }
          Interfaces using crypto map R1-R2_MAP:
                  Serial0/0/0
```

19.5 IPsec VPN

Send Interesting Traffic

Traffic from the LAN interface on R1 destined for the LAN interface on R2 is considered interesting traffic because it matches the ACLs configured on both routers. An extended ping from R1 will effectively test the VPN configuration.

```
R1# ping 192.168.1.1 source 10.0.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
Packet sent with a source address of 10.0.1.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/4 ms
R1#
```

Verify the ISAKMP and IPsec Tunnels

To verify that tunnels have been established, use the **show crypto isakmp sa** and **show crypto ipsec sa** (next slide) commands.

```
R1# show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
172.30.2.2      172.30.2.1      QM_IDLE           1005 ACTIVE

IPv6 Crypto ISAKMP SA

R1#
```

Verify the ISAKMP and IPsec Tunnels (Cont.)

The output for the ISAKMP tunnel is shown below. Notice that the tunnel is active between the two peers, 172.30.2.1 and 172.30.2.2.

```
R1# show crypto ipsec sa

interface: Serial0/0/0
    Crypto map tag: R1-R2_MAP, local addr 172.30.2.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.0.1.0/255.255.255.0/0/0)
   remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
   current_peer 172.30.2.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 4
    #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 172.30.2.1, remote crypto endpt.: 172.30.2.2
     plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
     current outbound spi: 0xD3E56A5F(3555027551)
     PFS (Y/N): Y, DH group: group24

     inbound esp sas:
      spi: 0x5D620493(1566704787)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2019, flow_id: Onboard VPN:19, sibling_flags 80004040, crypto map: R1-R2_MAP
        sa timing: remaining key lifetime (k/sec): (4155730/802)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE(ACTIVE)

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:
      spi: 0xD3E56A5F(3555027551)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2020, flow_id: Onboard VPN:20, sibling_flags 80004040, crypto map: R1-R2_MAP
        sa timing: remaining key lifetime (k/sec): (4155730/802)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE(ACTIVE)

     outbound ah sas:

     outbound pcp sas:
R1#
```

Video - Site-to-Site IPsec VPN Configuration

This video will demonstrate configuring a Site-to Site IPsec VPN Tunnel.

Packet Tracer - Configure and Verify a Site-to-Site IPsec VPN

In this Packet Tracer, you will complete the following objectives:

- Verify connectivity throughout the network
- Configure router R1 to support to site-to-site IPsec VPN with R3

Lab - Configuring a Site-to-Site VPN

In this lab, you will complete the following objectives:

- Configure basic device settings.
- Configure a site-to-site VPN using Cisco IOS.

19.6 Implement Site-to-Site IPsec VPNs with CLI Summary

What Did I Learn in this Module?

- IPsec negotiation to establish a VPN involves five steps, which include IKE Phase 1 and Phase 2.
- An ISAKMP tunnel is initiated when host A sends “interesting” traffic, defined by an ACL, to host B.
- IKE Phase 1 then begins and the peers negotiate the ISAKMP SA policy.
- IKE Phase 2 begins and the IPsec peers use the authenticated secure tunnel to negotiate the IPsec SA policy.
- The IPsec tunnel is created, and data is transferred between the IPsec peers based on the IPsec SAs.
- Implementing a site-to-site VPN requires configuring settings for both IKE Phase 1 and Phase 2.
- To enable multicast routing protocol traffic, the peers would need to be configured with a GRE tunnel.
- The ISAKMP policy lists the SAs that the router is willing to use to establish the IKE Phase 1 tunnel.
- Use the **show crypto isakmp default policy** command to view the default policies.

What Did I Learn in this Module?

- To configure a new ISAKMP policy, use the **crypto isakmp policy** command. The five SAs to configure are hash, authentication, group, lifetime, and encryption (HAGLE).
- Configure an ACL to define interesting traffic.
- Use the **crypto ipsec transform-set** command to configure the set of encryption and hashing algorithms that will be used to transform the data that is sent through the IPsec tunnel.
- To finish the configuration to meet the IPsec security policy you must bind the ACL and the transform set to the map, specify the peer’s IP address, configure the DH group, and configure the IPsec tunnel lifetime.
- Use the **show crypto map** command to verify the crypto map configuration.
- To apply the crypto map, enter interface configuration mode for the outbound interface and configure the **crypto map** *map-name* command.
- Test the two tunnels by sending interesting traffic across the link.
- To verify that tunnels have been established, use the **show crypto isakmp sa** and **show crypto ipsec sa** commands.

New Terms and Commands

- **ip access-list extended** *name*
    - **permit udp** *source wildcard destination wildcard* **eq isakmp**
    - **permit esp** *source wildcard destination wildcard*
    - **permit ahp** *source wildcard destination wildcard*
- **show crypto isakmp default policy**
- **crypto isakmp policy** *priority*
    - **encryption** *encryption-type*
    - **hash** *hash-algorithm*
    - **authentication** *auth-type*
    - **group 24** *dh-group*
    - **lifetime** *seconds*
- **crypto isakmp key** *keystring* **address** *peer-address*
- **crypto isakmp key** *keystring* **hostname** *peer-hostname*
- **show crypto isakmp sa**

New Terms and Commands (Cont.)

- **crypto ipsec transform-set** *name encryption algorithm*
- **crypto map** *map-name seq-num* { **ipsec-isakmp** | **ipsec-manual** }
- **show crypto map**
- **interface** *if-name*
    - **crypto map** *map-name*
- **show crypto ipsec sa**

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